Q) What is Measurement?

MEASUREMENT:- Can be said to be the process or act of finding the size, quantity or degree of something. It gives us a repeatable and dependable way of quantifying the world in which we live. It is essentially the act of comparing some unknown value with a value which is assumed to be know (otherwise known as a standard).

Q3 All measurement systems consist essentially of three parts; what are they?

i) THE SENSING ELEMENT:- This frequently called the transducer is the first element. It is in some way in contact with what is being measured and produces a signal which is related to the quantity being measured.

ii) THE SIGNAL CONVERTER:- The output of the sensing element then passes through a second element before reaching the display. This second element can take many forms. In general, it can be considered the signal converter in that the signal from the sensing element is converted into a form which is suitable for the display or control element.

(iii) THE DISPLAY :- The third part of the measuring system could be the display or a control system. The display element is where the output from the measuring system is displayed. ~~This may, for~~

(1c) Define Calibration ?

CALIBRATION :- may be defined, in general, as the process for determination, by measurement or comparison with a standard of the correct value each scale reading on a meter or measuring instrument; or determination the setting of a control device respond to particular value of volt, current, frequency, pressure, flow or some other output.

(2a) Define the following :- (i) Repeatability. (ii) Reproducibility (iii) Sensitivity. (iv) precision (v) Resolution (vi) Threshold.

(i) REPEATABILITY :- The repeatability of an instrument is it's ability to display the same reading for repeated applications of the same value of the quantity being measured.

(ii) REPRODUCIBILITY :- The reproducibility or stability of an instrument is it's ability to display the same reading when it is used to measure a constant quantity over a period of time or when that quantity is measured on a number of occasions.

i) SENSITIVITY :- This can be defined as the ratio of the change in output to the change in input which cause it, at steady state conditions. The sensitivity of an instrument is given by.

$$\text{Sensitivity} = \frac{\text{Change in instrument scale reading}}{\text{Change in the quantity being measured}}$$

ii) PRECISION :- This is the degree of exactness for which an instrument is designed to perform. It consists of two characteristics, conformity and the number of significant figures to which the measurement may be made.

iii) RESOLUTION :- The resolution or discrimination of an instrument is the smallest change in the quantity being measured that will produce an observable change in the reading of the instrument.

iv) THREHOLD :- If the quantity being measured is increased from zero a certain minimum level might have to be reached before the instrument respond and gives a detectable reading. This is called the threhold.

v) ACCURACY :- The accuracy of an instrument is the extent to which the reading it gives might be wrong or the ability of an device or a system to respond to the true value of a measured variable under reference condition.

(viii) RANGE :- The range of an Instrument is the limits between which reading can be made.

(Xi) DEAD SPACE :- The dead space of an Instrument is the range of values of the quantity being measured for which it gives no reading.

(X) HYSTERESIS :- Instrument can give different reading for the same value of measured quantity depending on whether that value has been reached by a continuously increasing change or continuously decreasing change.

(2B) Describe the principles, construction and working of thermistor as temperature measuring device?

THERMISTORS :- (Also known as "thermally sensitive resistors) are semi conductors made from a specific mixture of pure oxides of nickel, manganese, copper, cobalt, iron magnesium, titanium and other metals sintered at temperature above 982°C.

Thermistors are available in a number of configuration, most familiar is the bead type usually glass coated. They can also be made into washers, discs, or rods and can also be encapsulated in plastic, cemented, soldered in bolt, encased in glass tubes, needles, or a variety of other forms.

To measure temperature with a thermistor, It is placed in the environment whose temperature is to be measured. As the temperature of the substance or environment Increases, the resistance of the thermistor increase and vice versa. This change in thermistor resistance can be detect which will be the measured of the temperature of the substance. Generally, the thermistor is placed as one leg of a wheatstone bridge. At balance condition, when there is no change in temperature, the galvanometer Indicates zero. As the temperature Increases or decrease, the resistance of the thermistor also increase or decrease due to which the wheatstone bridge circuit becomes unbalanced. thus an electric current flows.

2c State three advantages and three disadvanges of thermistors?

3 ADVANTAGES

(i) Small size and fast response.
(ii) Suitability for narrow spans.
(iii) Low cost.

3 DISADVANTAGES

(i) Nonlinear temperature versus resistance curve.
(ii) unsuitable for wide temperature spans.
(iii) Problems due to Interchangeability of Individual element.
(iv) Doubtful stability at higher temperatures.
(v) Being limited for process application.
(vi) Too much resistance requiring shielded power lines, filter, or DC voltage.

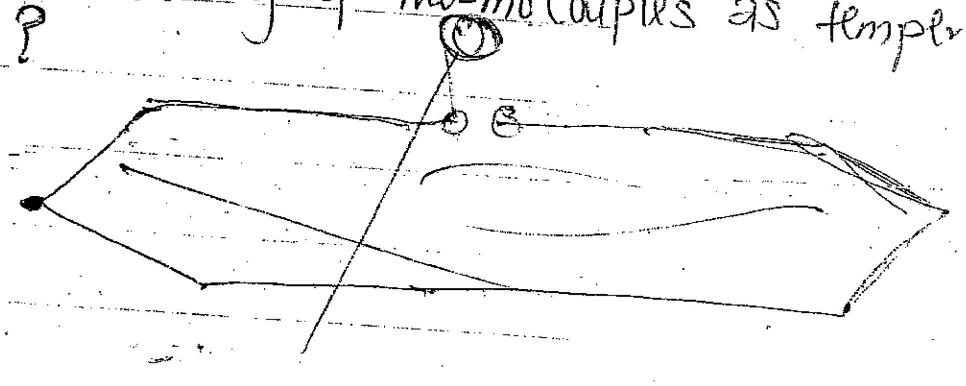
3a) What are the ~~three~~ way in which instrument can use for making measurement?

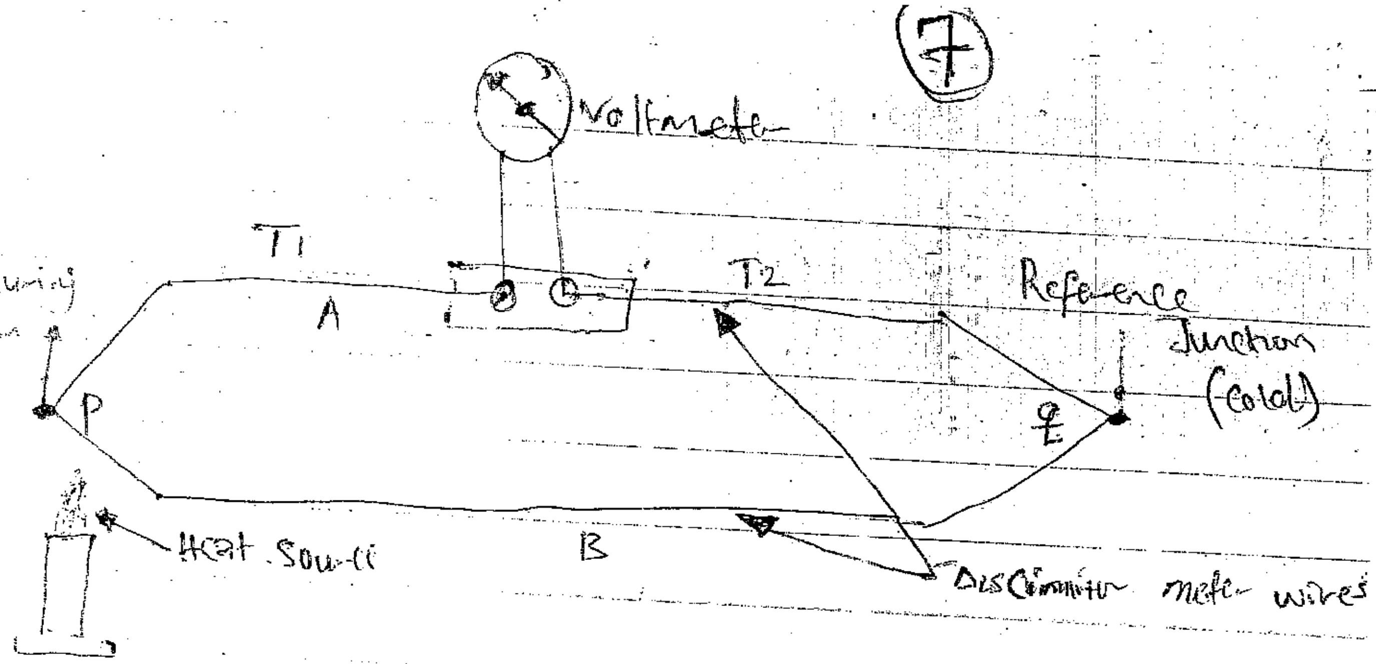
i) Obtaining data for some events or items:- This could for instance be the making out of an item for machining and involve measurement of length and angles.

ii) Inspecting an item to see if it matches the specification:- In production engineering measurement are often concerned with determining whether an item being produced has the right dimension, shape, electrical resistance etc. that have been specified for the item.

iii) Making measurement to ensure that a process is kept under control:- many industrial processes are continuous. The purpose of measurement in this case is to ensure the proper control of the process.

3B) Describe with the aid of a sketch, the principle, construction and working of thermocouples as temperature measuring devices?





THERMOCOUPLES:- The working principle of a thermocouple depends on the thermo-electric effect. If two dissimilar metals are joined together so as to form a close circuit there will be two junctions where they meet each other. If one of these junctions is heated then a current flow in the circuit which is detected by a galvanometer. The amount of current produced depends on the difference in temperature of two junction and the characteristics of the two metals. This was first observed by Seeback in 1821 and is know as Seeback effect. Instrument which record the variations in current flow are calibrated in term of temperature and are known ~~of metal~~ as thermocouples pyrometers.

3a States three advantages and three disadvantages of thermocouples?

ADVANTAGES

(i) Rugged Construction. (ii) Inexpensive. (iii) Good accuracy (iv) Long transmission distances (v) Good reproducibility (vi) Calibration checking made easy. (vii) Extremely wide temperature range from −270 to 2800°C.

DISADVANTAGES

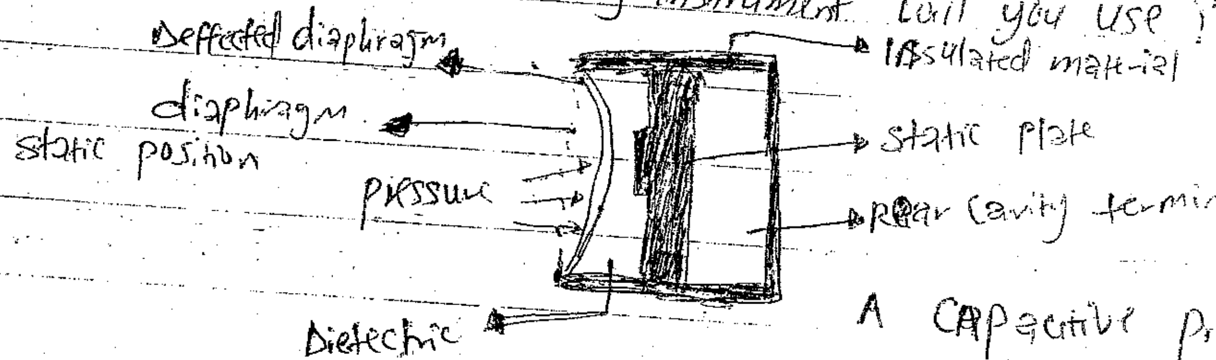
(i) limited use in temperature spans lee than about 33° because of the relatively small change in junction voltages wi temperature.

(ii) Non linearity of voltage-temperature relationship.

(iii) Require much of amplification for many applications.

(iv) Need for expensive accessories for controls application

4A You are required to design a pressure measuring system that can withstand a lot of vibration and shock, is extremely sensitive, has a good frequency response, and can give rapid response to change in pressure which of the pressure measuring instruments will you use?



A Capacitive pressure transducer

3B) With the aid of a diagram explain the principle, construction and working of the instrument mentioned in (2B) above?

C) What are the disadvantages of the instrument?

i) The metallic parts of the capacitor must be insulated from each other.

ii) The performance of a capacitive transducer is adversely affected by dirt and other contaminants, because they change the ~~di~~ dielectric constant.

iii) Sensitivity of a capacitive transducer is adversely affected by changes in temperature.

5A) List the various methods used for inferential flow measurement?

i) Variable head or differential meter (ii) Variable area meters.
(iii) Magnetic meters (iv) Turbine meters. (v) Target meters.
(vi) Thermal meters (vii) vortex meters (viii) Ultrasonic flow meters.

5B) What are the advantages and disadvantes of Variable head flow measuring Instruments?

ADVANTAGES

(i) It is accurate and reliable.
(ii) It is adaptable to any pipe size and flow rate.
(iii) It can easily be removed without shutting down the process.
(iv) It cost is relatively low especially for large lines.
(v) It offer the widest applicational coverage of any type of matter.

DISADVANTAGES

(i) There is relatively high permanent pressure loss in the
(ii) It is difficult to measure pulsating flow with this flow meter.
(iii) It is difficult to use for slurry service.
(iv) Its accuracy is dependent on many fixed fluid characteristics such as temperature, pressure, compressibility e.t.c

5C) The periodic maintenance of pressure measuring Instrument Consists of Three things, What are they?

(i) Visual Inspection. (ii) Blowdown and Venting (iii) Cleaning and lubrication.

(i) Visual inspection:- any leaks or damage in piping and wiring of the instrument can be examined visually and small Corrections may be done during that period.

(11)

i) Blowdown and Venting :- blowdown remove dirt and foreign materials from Instrument line before they plug the line. by Venting gas trapped in the liquid filled line, gas build up can be avoided, which also cause faulty pressure reading.

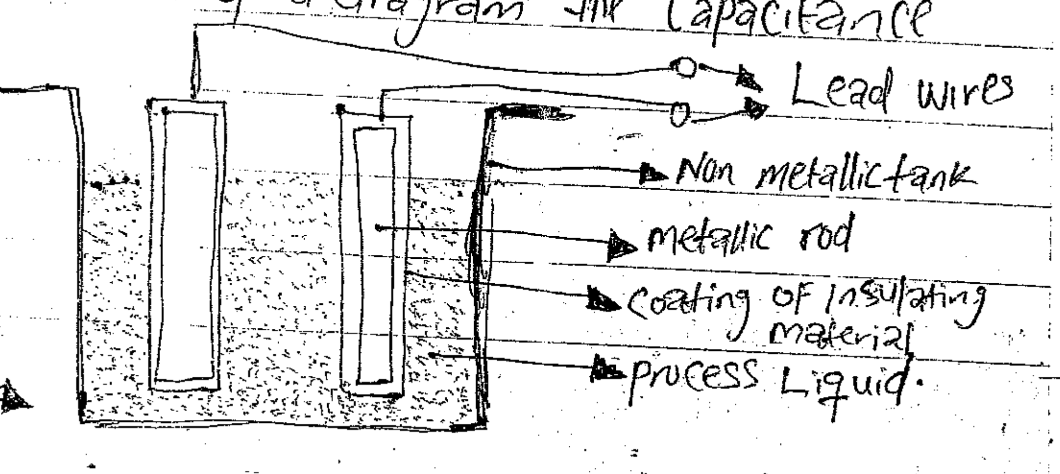
ii) Cleaning and lubrication :- Instrument with mechanical linkages subject to wear and misalignment. Dirt may clog the linkages, causing the mechanism to stick.

6a) List the hydrostatic pressure methods used for liquid level measurement?

i) Direct methods. (ii) Indirect methods.

i) DIRECT METHODS :- This is the simplest method of measuring liquid level. In this method, the liquid level is measured directly by means of the following level indicators:
Hook-type indicator, sight glass and float-type.
(ii) INDIRECT METHODS:- the following are the indirect methods of liquid level measurement generally used in industries; (i) Hydrostatic pressure type and Electrical methods.

6b) Describe with the aid of a diagram the Capacitance level indicator?



THE CAPACITANCE LEVEL INDICATOR

6c Write down the advantages and the disadvantages of the Capacitance level Indicator?

ADVANTAGES

i) It is very useful in a small system. ii) It is very sensitive. iii) It is suitable for continuous indication or control. iv) It is good for use with slurries. v) There are no moving parts exposed to the flu

vi) probe materials for most corrosive fluid are available.

vii) Remote adjustment of span and zero is possible in this type level Indicator.

DISADVANTAGES

i) it's sensitivity is adversely affected by change in temperature.

ii) Measured fluid must have proper dielectric qualities.

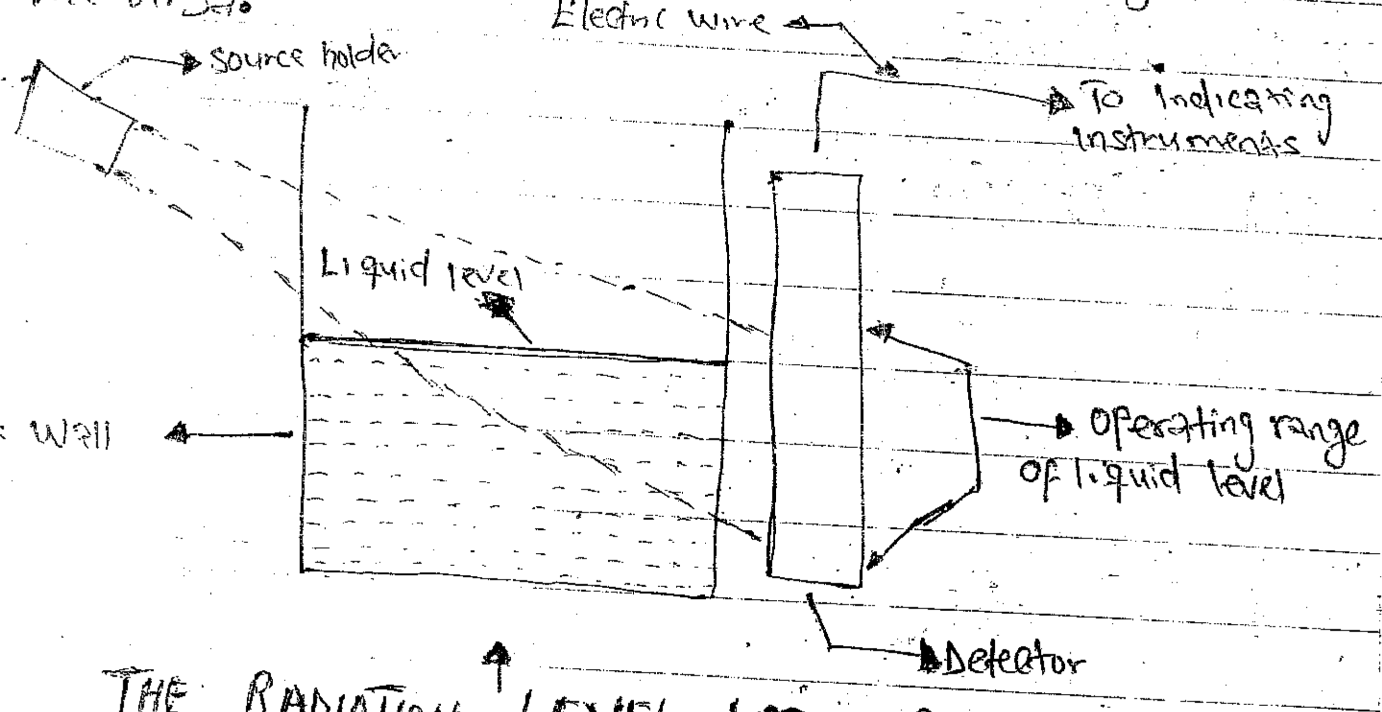
iii) They usually require recalibration if measured material change

iv) Probe length and mounting must suit the tank.

v) The performance of a capacitive level indicator is severely affected by dirt and other contaminants, because they chan the dielectric constant.

1A: Briefly describe the construction and working of a radiation level detector?

Construction and working :- This consist of a gamma rat holder on one side of the tank and a gamma detector on the other side of the tank. The gamma rays from the source are directed towards the detector in a thin band of radiation. When the gamma rays penetrate the thick walls of the tank, its energy level afterwards is greatly reduced. The radiation received at the gamma detector is inversely proportional to the thickness of the tank walls and the medium between the radiation source and the detector. That is the thicker the medium between source and detector, the less the radiation received by the detector and vice versa.



THE RADIATION LEVEL DETECTOR

6B) What are the advantages and disadvantages of radiation level detector?

ADVANTAGES

(i) There is no physical contact with the liquid.
(ii) They are suitable for molten metal as well as liquid of all types.
(iii) They are useful at very high temperature and pressure.
(iv) They have good accuracy and response.
(v) They have no moving parts.

DISADVANTAGES

(i) Reading is affected by change in liquid density.
(ii) Radiation source holder may be heavy.
(iii) Their cost is relatively high.

Q2B CONSTRUCTION OF THE THERMISTOR